

Test Plan & Test Report

Smart Biometric Bridge (SBB) — authenticated punch-ingest path

Field	Detail
Document	Test Plan & Test Report
System under test	SmartPRS by Ametecs — SBB ingest path (/api/v1)
Test file	tests/Feature/SbbIngestTest.php — 23 automated cases
Framework	Pest 3 on PHPUnit (pestphp/pest ^3.0)
Database under test	In-memory SQLite — the live MySQL database is never touched
Prepared	16 August 2026
Automated run status	NOT YET EXECUTED — see Section 2
Companion document	SmartPRS — Session Report (SBB ingest path)

Test isolation — safe to run at any time

phpunit.xml sets DB_CONNECTION=sqlite and DB_DATABASE=:memory:, and Feature tests use RefreshDatabase. Every test builds a fresh throwaway database in RAM and discards it.

Your live MySQL smartprs data is never read, written, or migrated by running these tests.

MAIL_MAILER=array also means no email leaves the machine.

Contents

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1. Scope and objectives

1.1 What is being tested

The authenticated JSON punch-ingest path added for Smart Biometric Bridge:

- GET /api/v1/ping — health and identity check
- POST /api/v1/attendance/punches — batch punch ingest
- App\Http\Middleware\ApiKeyAuth — key authentication and scope enforcement
- App\Services\PunchIngestService — time parsing, employee matching, de-duplication, quarantine, replay
- BiometricConfigController::mapEmployee — the replay trigger
- The three new migrations and their unique indexes

1.2 What must NOT break (regression scope)

- The /iclock ADMS endpoints and PushController — real ZKTeco hardware depends on them
- ETimeOfficeService::import and the three pull-based connectors, which now write against two new unique indexes
- All existing attendance_logs rows and the reports and payroll logic that read them

1.3 Acceptance criteria

Definition of done

All 23 automated cases pass, and the full existing suite still passes.

The five manual smoke tests in Section 5 pass against a real server after migration.

If any case fails, the migration must not be applied to production.

2. Execution status

The automated suite has been written but not yet executed

The suite was authored in a cloud environment with no network access to Packagist, GitHub, or any Composer mirror, so no Laravel test harness could be built there. That environment also had no shell access to the Windows development machine.

A one-click runner, `RUN-SBB-TESTS.bat`, has been placed in `C:\laragon\www\smartprs2`. It runs both suites and writes the full output to `sbb-test-results.txt`.

Until that runner is executed and its output reviewed, the "Result" column in Section 4 is unfilled and the acceptance criteria in 1.3 are unmet.

2.1 How to execute

Option A — double-click:

```
C:\laragon\www\smartprs2\RUN-SBB-TESTS.bat
```

Option B — from a Laragon terminal:

```
cd C:\laragon\www\smartprs2
vendor\bin\pest tests/Feature/SbbIngestTest.php
vendor\bin\pest
```

2.2 Verification already completed

Six checks were carried out directly, independently of the automated suite. These are evidence, not substitutes for it.

#	Check	Method	Result
V1	PHP syntax, all 17 files	<code>php -l</code> on every file	PASS — no syntax errors
V2	Blade directive balance	Directive count on the two new views	PASS — 8/8 <code>@if</code> , 2/2 <code>@foreach</code> , 1/1 <code>@section</code>
V3	Naive-local time parsing rules	Parser extracted; 15 inputs x 3 timezones	PASS — 45/45 cases
V4	Unique-index and dedupe semantics	Post-migration schema built in real SQLite	PASS — see 2.4
V5	PHP 8.2 compatibility for <code>smartprs-l3</code>	Syntax review against 8.2 feature set	PASS — no 8.3-only syntax
V6	On-prem route blocking	<code>Edition::BLOCK_SAAS</code> pattern match	PASS — new routes not blocked

2.3 V3 evidence — time parsing, 45 of 45 passing

Input	Expected	Actual
2026-08-16 09:41:00	accept, unchanged	accepted, unchanged
2026-08-16 09:41:00+05:30	reject TIME_FORMAT	rejected
2026-08-16 09:41:00-08:00	reject TIME_FORMAT	rejected
2026-08-16T09:41:00Z	reject TIME_FORMAT	rejected
2026-08-16T09:41:00+0530	reject TIME_FORMAT	rejected
2026-08-16 09:41:00 UTC	reject TIME_FORMAT	rejected
2026-08-16 09:41:00 IST	reject TIME_FORMAT	rejected
16/08/2026 09:41:00	reject TIME_FORMAT	rejected
2026-02-30 09:41:00	reject — invalid date rolls over	rejected
2026-13-01 09:41:00	reject TIME_FORMAT	rejected
2026-08-16 25:00:00	reject TIME_FORMAT	rejected
2026-08-16 09:41	reject — seconds missing	rejected
(empty string)	reject TIME_FORMAT	rejected
2026-12-31 23:59:59	accept, unchanged	accepted, unchanged
2026-01-01 00:00:00	accept, unchanged	accepted, unchanged

Each input was run under Asia/Kolkata, UTC and America/New_York. Behaviour was identical in all three, which is the point: the timezone never changes how the digits are read.

2.4 V4 evidence — dedupe and unique indexes in real SQLite

A table matching the post-migration schema was created, seeded with legacy rows including a duplicate pair, and exercised. Verbatim output:

```
rows before dedupe: 3
index refused while duplicates exist (as designed)
rows after dedupe: 2
both unique indexes created OK
legacy rows with NULL external_id coexist: 4
1st send -> inserted=1 (accepted)
identical re-send -> inserted=0 (duplicate)
same moment, new ext id -> inserted=0 (duplicate)
same ext id, other tenant -> inserted=1 (accepted)
legacy updateOrInsert -> updated=1 (no constraint violation)
EMP1043 in tenant 1: 1
```

Line	What it proves
index refused while duplicates exist	The migration MUST run attendance:dedupe first. This is why the ordering is not optional.
legacy rows with NULL external_id	Existing history is unaffected — NULL never collides in a unique index.

Line	What it proves
coexist: 4	
identical re-send -> inserted=0	The sender's idempotency key works. A retry is a no-op.
same moment, new ext id -> inserted=0	The natural key catches a duplicate even when the sender generates a fresh id.
same ext id, other tenant -> inserted=1	Tenants are independent. One customer's ids cannot block another's.
legacy updateOrInsert -> updated=1	The old import path still works against the new indexes — no regression.

3. Test environment

Setting	Value	Source
APP_ENV	testing	phpunit.xml
DB_CONNECTION	sqlite	phpunit.xml
DB_DATABASE	:memory:	phpunit.xml
MAIL_MAILER	array — nothing is sent	phpunit.xml
QUEUE_CONNECTION	sync — queued jobs run inline	phpunit.xml
SESSION_DRIVER	array	phpunit.xml
CACHE_STORE	array	phpunit.xml
Database lifecycle	RefreshDatabase — fresh per test	tests/Pest.php
Test runner	Pest 3 / PHPUnit	composer.json
PHP	8.3 (smartprs2) / 8.2 (smartprs-l3)	composer.json

3.1 Shared fixtures

Helper	Defined in	Purpose
makeTenantCompany()	tests/Pest.php (existing)	Creates a tenant and its first company
makeEmployee()	tests/Pest.php (existing)	Creates an active employee with a CTC
makeUser()	tests/Pest.php (existing)	Creates a user with a Spatie role
makeApiKey()	SbbIngestTest.php (new)	Issues a key; returns the plaintext secret and its id
punch()	SbbIngestTest.php (new)	One well-formed punch, overridable field by field
postPunches()	SbbIngestTest.php (new)	POSTs a batch with the X-API-Key header

4. Automated test cases

23 cases. Case TC-08 is parameterised over 6 timestamp formats, so 28 assertions-groups execute in total. Fill the **Result** column when the runner has been executed.

TC-01 — ping identifies the customer so an installer can confirm the right client

Field	Detail
Area	Health check
Steps	Create tenant + company "Test Co". Issue a key scoped [ingest]. GET /api/v1/ping with X-Api-Key.
Expected result	HTTP 200. ok=true, app="SmartPRS", tenant_id matches the key, company_name="Test Co", scopes=["ingest"], timezone matches config, server_time is a non-empty ISO 8601 string.
Result (pass / fail)	

TC-02 — ingests a punch for a mapped employee and stores it in attendance_logs

Field	Detail
Area	Happy path
Steps	Create employee EMP1043 with device_user_id=1043. POST one punch (device 1043, 2026-08-16 09:41:00, IN, FINGERPRINT).
Expected result	HTTP 200, batch.accepted=1, results[0].status="accepted". Row exists with emp_code=EMP1043, correct tenant_id, source="sbb", direction="in", device_sn, device_user_id, verify_mode all stored, and punch_at exactly 2026-08-16 09:41:00.
Result (pass / fail)	

TC-03 — accepts an external_id once and calls the re-send a duplicate, leaving exactly one row

Field	Detail
Area	Idempotency
Steps	POST the identical punch payload twice.
Expected result	First: accepted. Second: batch.accepted=0, batch.duplicates=1, status="duplicate". Exactly ONE row in attendance_logs.
Result (pass / fail)	

TC-04 — treats the same employee, moment and source as one punch even under a fresh external_id

Field	Detail
Area	Idempotency
Steps	POST two punches with the same employee, timestamp and source but different external_id values.

Field	Detail
Expected result	First accepted, second "duplicate". Exactly ONE row — the natural key catches it.
Result (pass / fail)	

TC-05 — holds a punch for an unmapped PIN instead of discarding it, and says so

Field	Detail
Area	Quarantine
Steps	POST a punch for device_user_id=9999, which matches no employee.
Expected result	HTTP 200, batch.pending=1, batch.accepted=0, status="pending", reason="EMPLOYEE_NOT_MAPPED". A row exists in attendance_pending with resolved_at NULL. attendance_logs is empty — nothing written under a wrong identity.
Result (pass / fail)	

TC-06 — releases held punches into attendance when the device ID is finally mapped

Field	Detail
Area	Replay
Steps	POST a punch for unmapped PIN 9999 (goes to pending). Create employee EMP9999. Sign in as admin and POST /app/biometric-config/map with device_id=9999, emp_code=EMP9999.
Expected result	Map returns ok=true and promoted=1. The punch now exists in attendance_logs against EMP9999, with punch_at unchanged at 2026-08-16 09:41:00. The pending row has resolved_at stamped, and no unresolved rows remain — so it cannot be replayed twice.
Result (pass / fail)	

TC-07 — rejects a punch_at carrying a timezone offset rather than guessing what it meant

Field	Detail
Area	Time handling
Steps	POST a punch with punch_at = "2026-08-16 09:41:00+05:30".
Expected result	HTTP 200, batch.rejected=1, batch.accepted=0, status="rejected", reason="TIME_FORMAT". Nothing written to attendance_logs.
Result (pass / fail)	

TC-08 — rejects other zone-bearing and malformed timestamps (6 variants)

Field	Detail
Area	Time handling
Steps	POST a punch for each of: 2026-08-16T09:41:00Z, 2026-08-16T09:41:00+05:30, "2026-08-16 09:41:00 UTC", 16/08/2026 09:41:00, 2026-02-30 09:41:00, 2026-08-16 09:41.

Field	Detail
Expected result	All six rejected with reason TIME_FORMAT. attendance_logs remains empty in every case.
Result (pass / fail)	

TC-09 — stores the wall clock literally, never shifting it into UTC

Field	Detail
Area	Time handling
Steps	Set app timezone to Asia/Kolkata. POST a punch at 2026-08-16 09:41:00.
Expected result	Stored punch_at is exactly 2026-08-16 09:41:00. It is NOT 04:11 — the UTC rendering that the old path would have produced.
Result (pass / fail)	

TC-10 — cannot write a punch attributed to another tenant

Field	Detail
Area	Tenant isolation
Steps	Create tenants A and B, both with an employee EMP001. Map B's employee to device 1043. Using tenant A's key, POST a punch with employee_code=EMP001.
Expected result	Accepted, but written to tenant A. Tenant B has zero rows. The cross-customer match that the /iclock path allows does not occur.
Result (pass / fail)	

TC-11 — ignores a tenant_id supplied in the request body

Field	Detail
Area	Tenant isolation
Steps	Using tenant A's key, POST a body containing tenant_id=B and company_id=9999 alongside the punches array.
Expected result	Accepted and written to tenant A. The body values are discarded entirely. Tenant B has zero rows.
Result (pass / fail)	

TC-12 — holds a punch for an employee who belongs to a different tenant

Field	Detail
Area	Tenant isolation
Steps	Only tenant B has EMP1043 (mapped to device 1043). POST with tenant A's key.
Expected result	Status "pending" — not a match. attendance_logs empty; one row quarantined under tenant A.
Result (pass / fail)	

TC-13 — rejects a missing key with 401 and the documented body

Field	Detail
Area	Authentication
Steps	POST without any X-API-Key or Authorization header.
Expected result	HTTP 401 with body {"error":{"code":"API_KEY_401","message":"..."}} and a non-empty message.
Result (pass / fail)	

TC-14 — rejects an unknown key with 401

Field	Detail
Area	Authentication
Steps	POST with a well-formed but non-existent key.
Expected result	HTTP 401, error.code = API_KEY_401.
Result (pass / fail)	

TC-15 — rejects an expired key with 401

Field	Detail
Area	Authentication
Steps	Issue a key with expires_at set to yesterday. POST with it.
Expected result	HTTP 401, error.code = API_KEY_401.
Result (pass / fail)	

TC-16 — rejects a revoked key with 401

Field	Detail
Area	Authentication
Steps	Issue a key with active=false. POST with it.
Expected result	HTTP 401, error.code = API_KEY_401.
Result (pass / fail)	

TC-17 — rejects a key without the ingest scope with 403

Field	Detail
Area	Authorisation
Steps	Issue a key scoped [read] only. POST a punch with it, then GET /api/v1/ping with the same key.
Expected result	POST returns HTTP 403 with error.code = API_KEY_403 and a message naming "ingest". The same key still succeeds on ping, which requires authentication but no scope.

Field	Detail
Result (pass / fail)	

TC-18 — accepts the key as an Authorization Bearer header too

Field	Detail
Area	Authentication
Steps	GET /api/v1/ping with "Authorization: Bearer <key>" and no X-API-Key.
Expected result	HTTP 200 with the correct tenant_id.
Result (pass / fail)	

TC-19 — records last_used_at when a key is used

Field	Detail
Area	Telemetry
Steps	Confirm last_used_at is NULL on a fresh key. Call ping. Re-read the row.
Expected result	last_used_at is no longer NULL.
Result (pass / fail)	

TC-20 — reports a verdict for every punch in a mixed batch and never silently drops one

Field	Detail
Area	Batch behaviour
Steps	POST a batch of four: one valid, one same-moment duplicate, one unmapped PIN, one with a +05:30 offset.
Expected result	batch = received 4, accepted 1, duplicates 1, pending 1, rejected 1. results has exactly 4 entries, and each external_id carries its own correct status.
Result (pass / fail)	

TC-21 — rejects a malformed punch individually instead of failing the whole batch

Field	Detail
Area	Batch behaviour
Steps	POST a batch of two: one valid, one with direction="SIDEWAYS".
Expected result	batch.accepted=1, batch.rejected=1. The valid punch is stored — a bad neighbour does not cost it.
Result (pass / fail)	

TC-22 — rejects an empty or oversized batch with 422

Field	Detail
Area	Validation
Steps	POST punches=[] and then punches=[1001 items].
Expected result	Both return HTTP 422 with error.code = VALIDATION_422.
Result (pass / fail)	

TC-23 — leaves the unauthenticated ZKTeco push endpoint working

Field	Detail
Area	Regression
Steps	GET /iclock/cdata?SN=TESTSN001&options=all, then POST a tab-delimited ATTLOG body to /iclock/cdata?SN=TESTSN001&table=ATTLOG.
Expected result	The handshake returns 200 containing "GET OPTION FROM: TESTSN001". The ATTLOG upload returns 200. Real hardware behaviour is unchanged by the new indexes.
Result (pass / fail)	

5. Manual smoke tests (after migration, on a real server)

Run these once the migration has been applied, before re-enabling the biometric feed.

MT-01 — Create a key

Settings → API Keys → Create key, name "SBB — <site>", scope ingest. **Expected:** the secret is displayed exactly once, in a green panel, and the list shows the prefix, scopes, "never" for last used, and status "active".

Result (pass / fail)	Notes

MT-02 — Confirm the right customer

```
curl -H "X-API-Key: sk_prs_xxxx_..." https://your-server/api/v1/ping
```

Expected: 200, and company_name is the customer you expect. If it names a different customer, stop — the key is scoped wrongly.

Result (pass / fail)	Notes

MT-03 — Send one punch

```
curl -X POST https://your-server/api/v1/attendance/punches ^
-H "X-API-Key: sk_prs_xxxx_..." ^
-H "Content-Type: application/json" ^
-d '{"punches":[{"external_id\":"smoke-test-001\","
  "device_sn\":"AXK7231900123\","device_user_id\":"1043\","
  "punch_at\":"2026-08-16 09:41:00\","direction\":"IN\","
  "verify_mode\":"FINGERPRINT\"}]}'
```

Expected: results[0].status is accepted (or pending if PIN 1043 is not mapped — which is also correct behaviour).

Result (pass / fail)	Notes

MT-04 — Send the SAME payload again

Expected: results[0].status is duplicate, and the row count in the database stays at 1.

```
SELECT emp_code, punch_at, direction, source, external_id
FROM attendance_logs WHERE external_id = 'smoke-test-001';
-- exactly ONE row; punch_at exactly 2026-08-16 09:41:00
```

Result (pass / fail)	Notes

MT-05 — The old hardware path still works

```
curl "https://your-server/iclock/cdata?SN=<a-real-serial>&options=all"
```

Expected: 200, with the option block beginning GET OPTION FROM: <serial>. Then confirm a real device uploads a punch as it did before.

Result (pass / fail)	Notes

6. Risks and things to watch

#	Risk	Likelihood	Mitigation
R1	attendance:dedupe deletes real rows that were not actually duplicates	Low	Run with --dry first and read the list. Take a database dump before running it. migrate:rollback will NOT restore deleted rows — the dump is the only way back.
R2	Index creation locks attendance_logs for minutes on a large table	Medium	Run in a maintenance window with the biometric feed paused.
R3	The direction column change (4 to 8) fails on an unusual driver	Low	Wrapped in try/catch; the migration continues. Only the value "unknown" needs the extra width.
R4	A key issued without a tenant cannot ingest	Low	By design — see Session Report 4.2. The UI stamps the creating user's tenant, and on-prem installs always have a tenant row.
R5	SBB does not send employee_code, and the device has no emp_prefix set	Medium	Punches land as "pending" rather than being lost, and appear in Settings → Pending Punches for mapping. Confirm emp_prefix on each device row before go-live.
R6	The two Settings screens are not reachable from the live SPA	High	They exist at /app/api-keys and /app/pending-punches and work by direct URL. Adding the menu entries to ApplicationController.php is an open item.

7. Result summary and sign-off

7.1 Automated run

Item	Value
Date and time of run	
Run by	
Environment / folder	
SBB suite — tests passed	
SBB suite — tests failed	
Full suite — result	
Output file attached	sbb-test-results.txt

7.2 Failures observed

Case	Failure detail	Action taken

7.3 Approval to migrate

Migration to production may proceed only when all automated cases pass and the manual smoke tests in Section 5 have been completed on a staging or freshly backed-up system.

Role	Name	Signature	Date
Developer / tester			
Reviewed by			
Approved for production	Ejaz Hussain		

Prepared 16 August 2026 — SmartPRS by Ametecs.